

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.4: Matrix multiplication and map composition

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Matrix product formulas

Definition: Matrix-matrix product

Let A be an $m \times n$ matrix, let B be an $n \times k$ matrix, and write

$$B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_k]$$

where $\mathbf{b}_j$ is column j of B . The product AB is the $m \times k$ matrix

$$AB = [A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_k].$$

Activity: Computing a matrix product using columns (U1-LO6)

Let

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ -2 & -6 \\ -1 & 0 \end{bmatrix}.$$

Compute AB .

Theorem

If A is $m \times n$ and B is $n \times k$, then the (i, j) -entry of AB is the dot product of row i of A with column j of B :

$$\begin{aligned}(AB)_{ij} &= \sum_{\ell=1}^n a_{i\ell} b_{\ell j} \\ &= a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}.\end{aligned}$$

Why is this true?

By the definition of matrix multiplication, column j of AB is $A\mathbf{b}_j$, where

$\mathbf{b}_j = (b_{1j}, \dots, b_{nj})$ is column j of B . The row dot-product rule for matrix-vector multiplication in Row dot-product view therefore gives

$$\begin{aligned}(AB)_{ij} &= (A\mathbf{b}_j)_i \\ &= \text{row}_i(A) \cdot \mathbf{b}_j \\ &= \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.\end{aligned}$$

Note: Row-column interpretation

The sum above is the dot product of row i of A and column j of B . Viewing both as n -vectors,

$$(AB)_{ij} = (a_{i1}, \dots, a_{in}) \cdot (b_{1j}, \dots, b_{nj}).$$

Pair entries in corresponding positions, multiply, and add.

Activity: Computing a matrix entry (U1-LO6)

If

$$A = \begin{bmatrix} -3 & -4 & 1 \\ 2 & 4 & 0 \\ 1 & -4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & -4 \\ -4 & -3 & -1 \\ 4 & 3 & 1 \end{bmatrix},$$

what is the $(2, 3)$ -entry of AB ?

Remark: Matrix-vector products as matrix products

A vector $\mathbf{x} \in \mathbb{R}^n$ can be viewed as an $n \times 1$ matrix. With this convention, if A is an $m \times n$ matrix, the matrix-vector product $A\mathbf{x}$ is a special case of matrix multiplication: its $m \times 1$ result represents a vector in $\mathbb{R}^m$, with entries given by the same row dot products.

Theorem: Properties of matrix multiplication

Assume that A, B, C are matrices of appropriate sizes and r is a scalar. Then

1. $A(BC) = (AB)C$.
2. $(A + B)C = AC + BC$.
3. $C(A + B) = CA + CB$.
4. $A(rB) = r(AB) = (rA)B$.
5. $(AB)^T = B^T A^T$.

Why is this true?

We prove property 5. Let A be $m \times n$ and B be $n \times k$. Then A^T is $n \times m$ and B^T is $k \times n$. Both $(AB)^T$ and $B^T A^T$ have shape $k \times m$. Transposition exchanges row and column indices, so the entry formula in the theorem gives

$$\begin{aligned}
((AB)^T)_{ij} &= (AB)_{ji} \\
&= \sum_{\ell=1}^n a_{j\ell} b_{\ell i} \\
&= \sum_{\ell=1}^n b_{\ell i} a_{j\ell} \\
&= \sum_{\ell=1}^n b_{i\ell}^T a_{\ell j}^T \\
&= \sum_{\ell=1}^n (B^T)_{i\ell} (A^T)_{\ell j} \\
&= (B^T A^T)_{ij}.
\end{aligned}$$

Here $b_{i\ell}^T$ and $a_{\ell j}^T$ denote the indicated entries of B^T and A^T , as in the definition. We can reverse the two scalar factors in each summand because real-number multiplication is commutative. Every corresponding entry agrees, so $(AB)^T = B^T A^T$.

Note: Shape habit for matrix products

Before forming AB , check the inner dimensions. If A is $m \times n$ and B is $n \times k$, then the shared inner dimension n makes the product possible, and the outer dimensions give AB shape $m \times k$.

Activity: Applying one matrix to many points (U1-LO5, U1-LO8)

Put the vertices of a shape in the columns of a matrix X . Then $Y = AX$ contains the transformed vertices.

```
import numpy as np

# Columns are vertices of the unit square.
# The final column repeats the first vertex to close the shape.
X = np.array([
    [0., 1., 1., 0., 0.],
    [0., 0., 1., 1., 0.]])
```

Activity: Applying one matrix to many points (U1-LO5, U1-LO8) (continued)

```
] )  
  
A = np.array([  
    [1., 1.],  
    [0., 1.],  
])  
  
Y = A @ X  
Y
```

Output:

```
array([[0., 1., 2., 1., 0.],  
       [0., 0., 1., 1., 0.]])
```

1. What are the columns of Y?
2. Which transformation from the gallery is this?
3. Why does the bottom edge stay fixed?

Warning

In this geometric example, the columns of X are points, so $Y = AX$ applies the same matrix to every point. In many data tables, observations are stored as rows. For example, a document matrix may have one document per row. Always check what rows and columns represent before interpreting a product.

A missing rule: AB need not equal BA

Multiplication of real numbers is commutative: $ab = ba$. Matrix multiplication is not commutative in general. For two square matrices A and B of the same size, we say that A and B commute when

$$AB = BA.$$

Some pairs commute and some do not.

Example: A product defined in only one order

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

The matrix A is 2×2 and B is 2×3 . The product AB is defined because the two columns of A match the two rows of B , and

$$AB = \begin{bmatrix} 1 & 2 & 8 \\ 0 & 1 & 3 \end{bmatrix}.$$

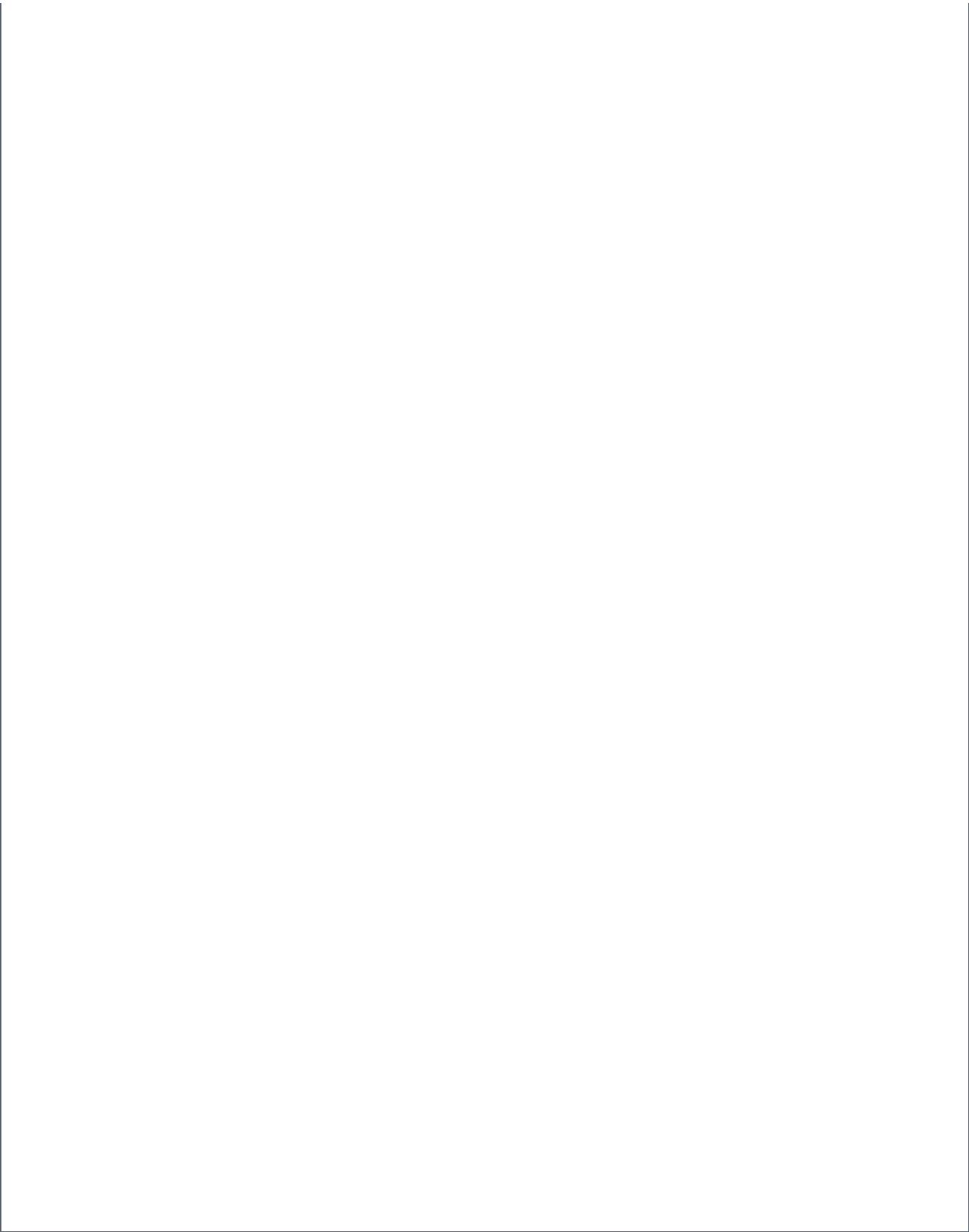
In the reverse order, B has three columns but A has only two rows, so BA is not defined. Thus knowing that AB exists does not guarantee that BA exists, let alone that the two products are equal.

Activity: Do these matrices commute? (U1-L06)

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Compute AB and BA . Do A and B commute?



Matrix multiplication as composition

If $T_A(\mathbf{x}) = A\mathbf{x}$ and $T_B(\mathbf{x}) = B\mathbf{x}$, then applying B first and then A gives

$$T_A(T_B(\mathbf{x})) = A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Matrix multiplication represents composition of matrix maps.

Activity: Composition shape check (U1-LO3, U1-LO6)

Suppose A is 4×3 and B is 3×2 . What is the shape of AB , and what is the domain and codomain of the composed map $\mathbf{x} \mapsto A(B\mathbf{x})$?

Activity: Commute or not? (U1-LO6)

For each pair below:

1. compute both products;
2. determine whether $AB = BA$;

Pair A: two distinct diagonal transformations. Use the horizontal stretch and reflection across the y -axis:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Pair B: a shared axis label is not enough. Use the same horizontal stretch and the horizontal shear:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Activity: Commute or not? (U1-LO6) (continued)

Pair C: reflection across the line $y = x$. Use the same horizontal stretch and

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Activity: Two horizontal shears (U1-LO6)

Let

$$H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The matrix G is a stronger horizontal shear.

1. Justify the claim that applying H twice has the same effect as applying G once.

Activity: Two horizontal shears (U1-LO6) (continued)

2. Compute HG and GH . Do the matrices commute?

Note

Some structural patterns settle the question quickly. Two diagonal matrices of the same size commute. For example, if

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \quad B = \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix},$$

then

$$AB = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix},$$
$$BA = \begin{bmatrix} ca & 0 \\ 0 & db \end{bmatrix} = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix} = AB.$$

The equality follows because multiplication of real numbers is commutative.

Every square matrix commutes with itself. The two horizontal shears in Two horizontal shears (U1-LO6) are closely related to this fact: as the

solution shows, $G = H^2$, so both HG and GH apply the same matrix H three times. Part 1 of Properties of matrix multiplication gives

$$HG = H(HH) = (HH)H = GH.$$

In general, compute AB and BA and compare.

Many token queries and keys

Example: Many token queries at once

Suppose a sequence has L tokens. Store the token vectors as rows of a matrix

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_L^T \end{bmatrix} \in \mathbb{R}^{L \times d}.$$

A transformer layer forms three views of the same token vectors:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The matrices W_Q, W_K, W_V are weight matrices. Their entries are adjusted during training. For now, treat them as ordinary matrices that assign each token three roles.

For destination position i , the vector $\mathbf{q}_i$ is its query: what that position uses to decide which source positions are relevant to its attention output.

For source position j , the vector $\mathbf{k}_j$ is its key: what is compared with the query from a destination position.

For source position j , the vector $\mathbf{v}_j$ is its value: the information that position can contribute after weights are chosen.

Write

$$Q = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix}, \quad K = \begin{bmatrix} \mathbf{k}_1^T \\ \mathbf{k}_2^T \\ \vdots \\ \mathbf{k}_L^T \end{bmatrix}.$$

Then

$$K^T = [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L].$$

So

$$QK^T = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix} [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L] = \begin{bmatrix} \mathbf{q}_1 \cdot \mathbf{k}_1 & \mathbf{q}_1 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_1 \cdot \mathbf{k}_L \\ \mathbf{q}_2 \cdot \mathbf{k}_1 & \mathbf{q}_2 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_2 \cdot \mathbf{k}_L \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_L \cdot \mathbf{k}_1 & \mathbf{q}_L \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_L \cdot \mathbf{k}_L \end{bmatrix}.$$

Thus the score matrix

$$S = QK^T$$

has entries

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Row i of S contains the scores used when computing the attention output at position i .

Activity: Many-token shape and entry check (U1-LO3, U1-LO6)

Suppose

$$X \in \mathbb{R}^{5 \times 4}$$

and

$$W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3}.$$

Let

Activity: Many-token shape and entry check (U1-LO3, U1-LO6) (continued)

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. Suppose

$$\mathbf{q}_2 = \begin{bmatrix} q_{21} \\ q_{22} \\ q_{23} \end{bmatrix}, \quad \mathbf{k}_4 = \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix}.$$

Expand the entry S_{24} .

5. What does S_{24} compare?
6. What does row 2 of S contain?

For this section, an

attention calculation

has three steps:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

The score S_{ij} compares the query at destination position i with the key at source position j . A larger score means source position j is more relevant to the attention output at destination position i .

The scores are not yet the output. For each token i , a weighting rule converts row i of S into numbers

$$\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iL}$$

that are nonnegative and add to 1. These are the

attention weights

for token i . They say how much token i

uses each value vector.

Let

$$A_{\text{att}} = [\alpha_{ij}]$$

be the attention-weight matrix. If the value vectors are stored as rows,

$$V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_L^T \end{bmatrix},$$

then the attention output is

$$H = A_{\text{att}} V.$$

Row i of H is

$$\mathbf{h}_i^T = \alpha_{i1} \mathbf{v}_1^T + \alpha_{i2} \mathbf{v}_2^T + \cdots + \alpha_{iL} \mathbf{v}_L^T.$$

Equivalently,

$$\mathbf{h}_i = \alpha_{i1} \mathbf{v}_1 + \alpha_{i2} \mathbf{v}_2 + \cdots + \alpha_{iL} \mathbf{v}_L.$$

Thus H is the attention-output matrix, and each row of H is a weighted average of value vectors. These rows contribute to later token representations; a full transformer layer combines them with the current representations and processes them further.

Activity: Many-token attention output (U1-LO3, U1-LO6, U1-LO8)

Suppose

$$X \in \mathbb{R}^{4 \times 3}, \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2}.$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A weighting step converts S into an attention-weight matrix

$$A_{\text{att}} \in \mathbb{R}^{4 \times 4}.$$

The attention output is

$$H = A_{\text{att}} V.$$

Activity: Many-token attention output (U1-LO3, U1-LO6, U1-LO8) (continued)

1. What are the shapes of Q , K , and V ?
2. What is the shape of $S = QK^T$?
3. What does S_{ij} compare?
4. What does row i of A_{att} tell you?
5. What is the shape of H ?
6. Why is each row of H a weighted average of value vectors?
7. Now suppose row 2 of A_{att} is

$$\alpha_2^T = [0 \quad 1/2 \quad 1/4 \quad 1/4]$$

and

$$V = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}.$$

Compute the attention-output vector $\mathbf{h}_2$.

Warning

In real transformer attention, the weighting rule usually uses scaled and masked scores before forming weights. For Unit 1, the important facts are: rows of A_{att} are weights, the weights add to 1, and $A_{\text{att}}V$ forms weighted averages of the rows of V .

Activity: The same shape check in code (U1-L03, U1-L08)

The following code performs a many-token shape check similar to the previous activity.

```
import numpy as np

X_tokens = np.ones((5, 4))
WQ = np.ones((4, 3))
WK = np.ones((4, 3))
WV = np.ones((4, 3))

Q = X_tokens @ WQ
K_tokens = X_tokens @ WK
V_tokens = X_tokens @ WV
scores_many = Q @ K_tokens.T

Q.shape, K_tokens.shape, V_tokens.shape, scores_many.shape
```

Output:

```
((5, 3), (5, 3), (5, 3), (5, 5))
```

1. Which line creates all query-key scores at once?
2. Why is `scores_many.shape` equal to `(5, 5)`?

